Engineering Mechanics 1

Week 6

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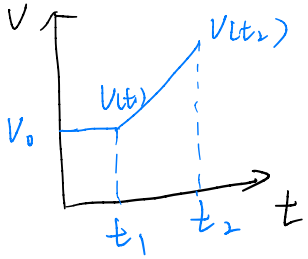
Autonomous Aerial Systems

Continuous loads - Differentials

Equilibrium:

→ system (group of beams and supports), subsystem (beam), subsections (section of beam)

→ equilibrium in subsections requires *internal forces* = stress resultants



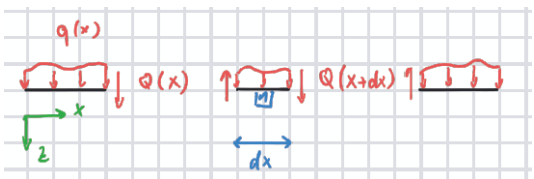
Stress resultants:

Normal force $N(x)' \quad \frac{\mathrm{d}N(x)}{\mathrm{d}x} = -n(x) \rightarrow N(x) = -\int_{x_a}^{x_b} n(x)\,dx + N_{x_a}$

Shear force $Q(x)' \quad \frac{\mathrm{d}Q(x)}{\mathrm{d}x} = -q(x) \rightarrow Q(x) = -\int_{x_a}^{x_b} q(x)\,dx + Q_{x_a}$

Bending moment $M(x)' \quad \frac{\mathrm{d}M(x)}{\mathrm{d}x} = Q(x) \rightarrow M(x) = \int_{x_a}^{x_b} Q(x)\,dx + M_{x_a} = -\int_{x_a}^{x_b}\left(\int_{x_a}^{x_b} q(x)dx + Q_{x_a}\right)dx + M_{x_a}$

Where do these formulas come from? *(see notes from lecture 8)*



For example, $Q(x) \rightarrow$ from static equilibrium in 1:

$\sum F = 0; \quad Q(x) - Q(x+dx) - q(x)dx = 0$

Assuming linearity, $\rightarrow Q(x+dx) = Q(x) + dQ(x)$, leading to

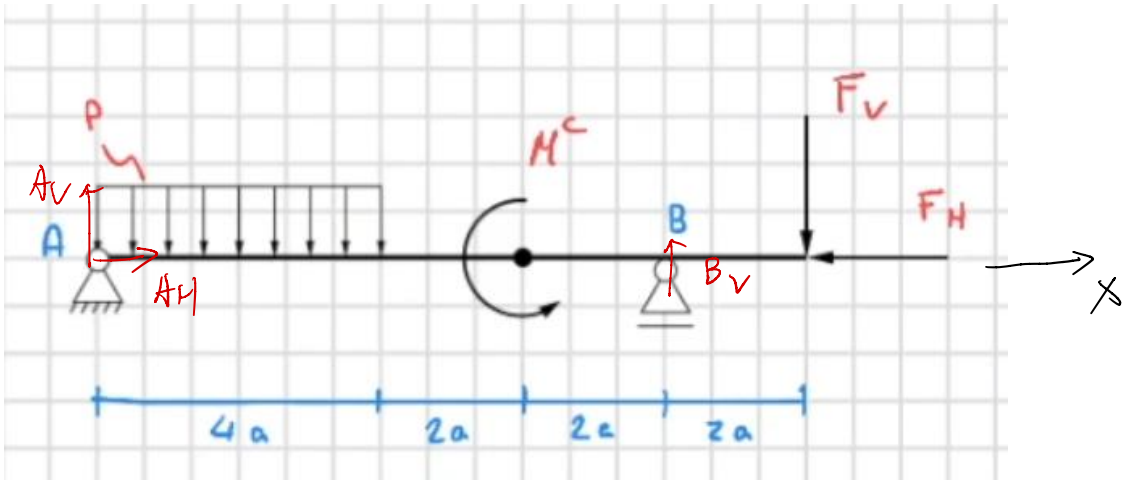
$$\frac{\mathrm{d}Q(x)}{\mathrm{d}x} = -q(x)$$

Boundary conditions

Symbol	Support	Known without computation	Support forces (known from global equilibrium)
	Pin	$M(x_0) = 0$	$N(x_0)$, $Q(x_0)$
	Parallel motion	$M(x_0) = 0$, $N(x_0) = 0$	$Q(x_0)$
	Fixed end	— — —	$N(x_0)$, $Q(x_0)$, $M(x_0)$
	Free end	$N(x_0) = 0$, $Q(x_0) = 0$, $M(x_0) = 0$	— — —

$q(x)$	< 0			$= 0$			> 0		
$Q(x)$	increasing			constant			decreasing		
	< 0	0	> 0	< 0	0	> 0	< 0	0	> 0
$M(x)$	left curved			straight			right curved		
	Decrea-sing	Min	Increa-sing	Decrea-sing	Horiz-ontal	Increa-sing	Decrea-sing	Max	Increa-sing

Exercise 1



We know

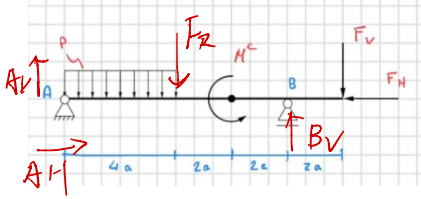
- P = 5 kN/m
- Mc = 4 kNm
- Fv = 5.2 kN
- Fh = 10 kN
- a = 1 m

Calculate

- Stress resultants, i.e., N,Q,M

Steps:
1> support reactions, A_V, B_V, A_H
2> stress resultants, N(x), Q(x), M(x)
3> Diagram
$\forall x \in [0..10a]$

① support reactions



$$\begin{cases} \Sigma F_H = 0 \Rightarrow A_H - F_H = 0 \\ \Sigma F_V = 0 \Rightarrow A_V - F_R + B_V - F_V = 0 \\ \curvearrowright \Sigma M_A = 0 \Rightarrow -2a \cdot F_R + M_C + 8a \cdot B_V - 10a F_V = 0 \end{cases} \quad , \quad F_R = 4aP$$

$$\begin{cases} A_H = F_H = 10\ kN \\ A_V = 14.2\ kN \\ B_V = 11\ kN \end{cases}$$

② stress resultants → $N(x)$, $Q(x)$, $M(x)$

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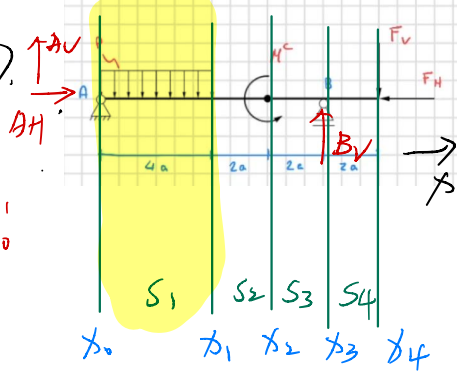
S_1

$N'_{x_0} = -A_H = -10\ KN$

$Q'_{x_0} = -(-A_V) = A_V = 14.2\ KN$

$M'_{x_0} = 0$





$N'_{x_1} = -\int_{x_0}^{x_1} n(x)\,dx + N'_{x_0} = -10\ [KN]$ constant

$Q'_{x_1} = -\int_{x_0}^{x_1} q(x)\,dx + Q'_{x_0} = -P x_1 + Q'_{x_0} = -5x_1 + 14.2\ [KN]$

Linear

$M'_{x_1} = \int_{x_0}^{x_1} Q(x)\,dx + M'_{x_0} = -\frac{5}{2}x_1^2 + 14.2x_1 + 0\ [KN\cdot m]$

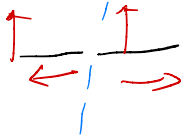
Quadratic

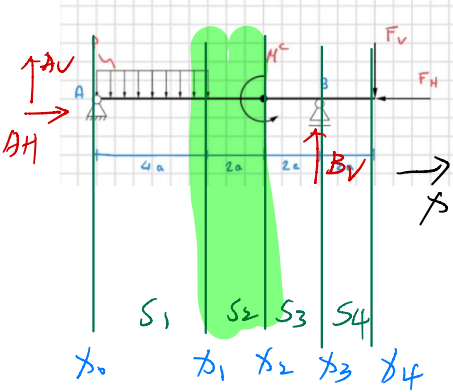
>> S_2

$N^2_{x_1} = N^1_{x_1} = -10$ kN

$Q^2_{x_1} = Q^1_{x_1} = -5.8$ kN

$M^2_{x_1} = M^1_{x_1} = 16.8$ kNm



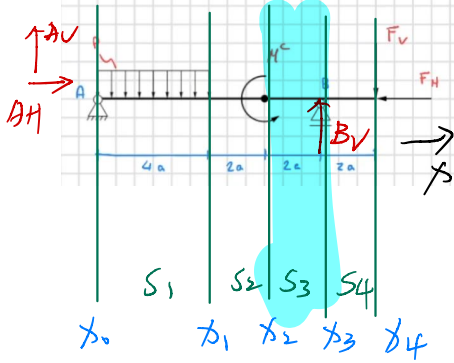


$$N^2_{x_2} = -\int_{x_1}^{x_2} n(x)dx + N^2_{x_1} = -10 \quad [kN]$$ constant

$$Q^2_{x_2} = -\int_{x_1}^{x_2} q(x)dx + Q^2_{x_1} = -5.8 \quad [kN]$$ constant

$$M^2_{x_2} = \int_{x_1}^{x_2} Q(x)dx + M^2_{x_1} = -5.8(x_2 - x_1) + 16.8 \quad [kN\cdot m]$$

linear



3) S_3

$$N^3_{x_2} = N^2_{x_2} = -10 \text{ KN}$$

$$Q^3_{x_2} = Q^2_{x_2} = -5.8 \text{ KN}$$

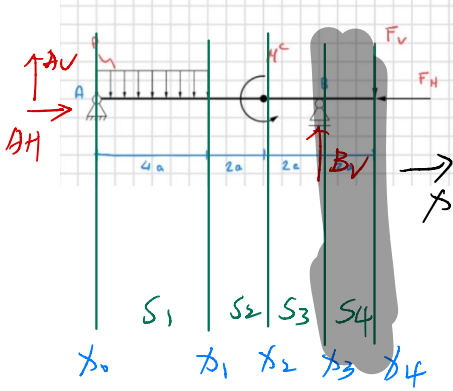
$$M^3_{x_2} = M^2_{x_2} - M_C = 1.2 \text{ KNm}$$

$$N^3_{x_3} = -\int_{x_2}^{x_3} n(x)\,dx + N^3_{x_2} = -10 \text{ [KN]} \quad \text{constant}$$

$$Q^3_{x_3} = -\int_{x_2}^{x_3} q(x)\,dx + Q^3_{x_2} = -5.8 \text{ [KN]} \quad \text{constant}$$

$$M^3_{x_3} = \int_{x_2}^{x_3} Q(x)\,dx + M^3_{x_2} = -5.8\,(x_3 - x_2) + 1.2 \text{ [KNm]}$$

linear



4) S_4

$$N_{x3}^4 = N_{x3}^3 = -10 \text{ KN}$$

$$Q_{x3}^4 = Q_{x3}^3 + B_V = 5.2 \text{ KN}$$

$$M_{x3}^4 = M_{x3}^3 = -10.4 \text{ KNm}$$

$$N_{x4}^4 = -\int_{x_3}^{x_4} n(x)\,dx + N_{x3}^4 = -10 \ [\text{KN}] \quad \text{constant}$$

$$Q_{x4}^4 = -\int_{x_3}^{x_4} q(x)\,dx + Q_{x3}^4 = 5.2 \ [\text{KN}] \quad \text{constant}$$

$$M_{x4}^4 = \int_{x_3}^{x_4} Q(x)\,dx + M_{x3}^4 = 5.2(x_4 - x_3) - 10.4 \ [\text{KNm}]$$

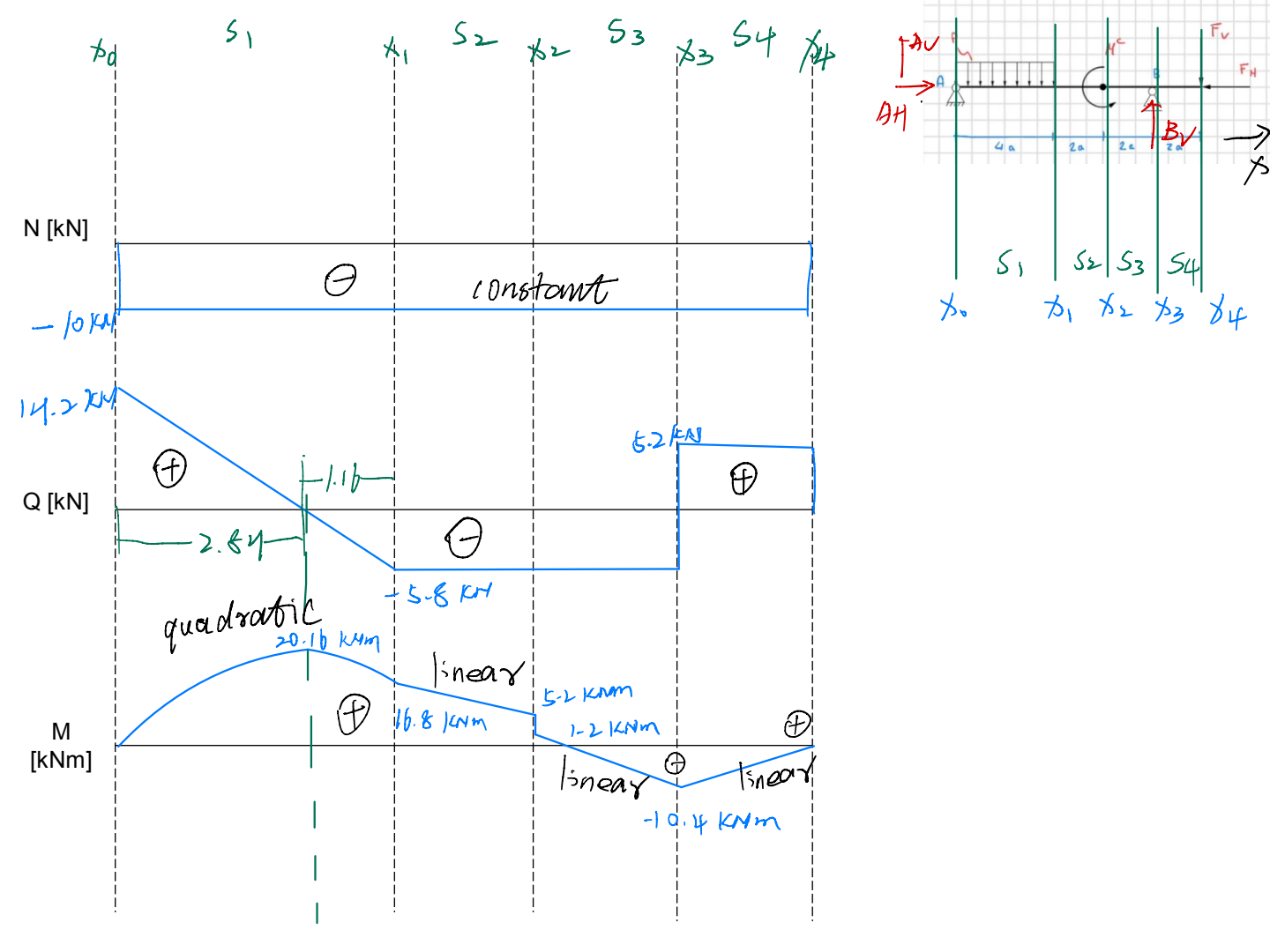
Linear

$\Rightarrow$

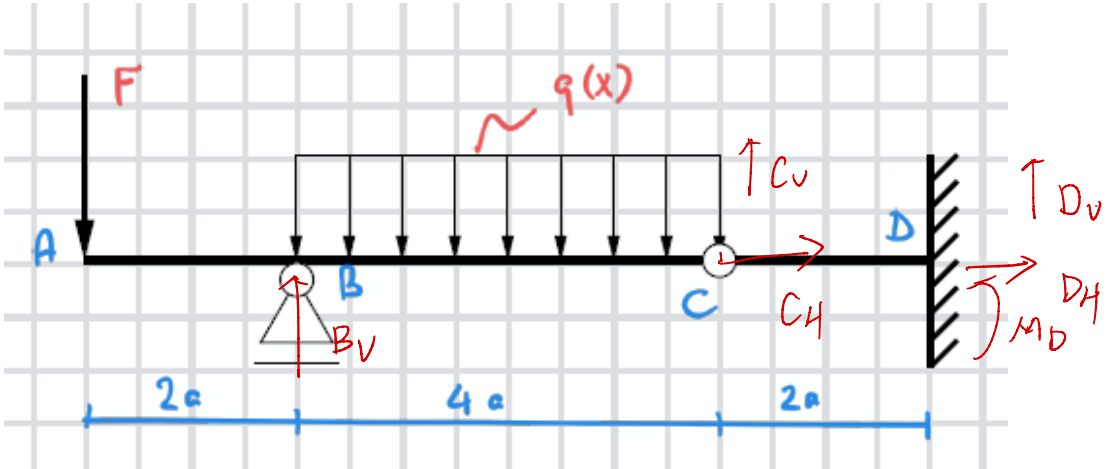
$$N_{x4}^4 = -F_H$$

$$Q_{x4}^4 = F_V \ (\text{opposite directions})$$

$$M_{x4}^4 = 0$$


S1
S2
S3
S4
x0
x1
x2
x3
x4
N [kN]
constant
−10 kN
Q [kN]
14.2 kN
1.1 l
2.8 l
−5.8 kN
5.2 kN
M [kNm]
quadratic
20.16 kNm
linear
16.8 kNm
5.2 kNm
1.2 kNm
linear
linear
−10.4 kNm
AV
AH
MC
Fv
FH
BV
4a
2a
2a
2a
x

Exercise 2



We know
- F = 6 kN
- q = 2 kN/m
- a = 1 m

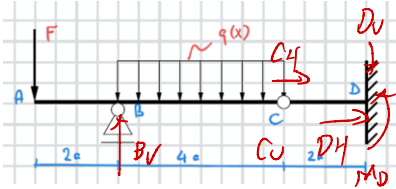
Calculate
- Stress resultants, i.e., N,Q,M

6teps:

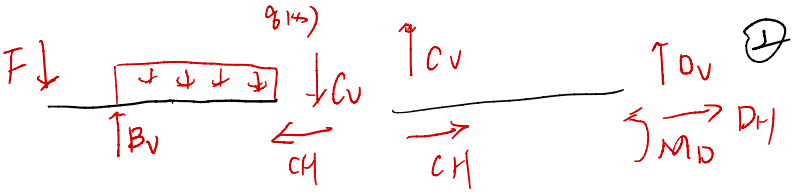
1> support and interal reactions : $B_V, C_V, C_H, D_V, D_H, M_D$

2> stress resultants → diagram : $N(x), Q(x), M(x)$

1) support and internal reactions



① ②

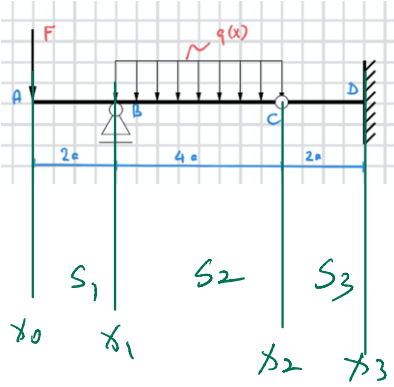


① $\begin{cases} \Sigma F_H = 0 \Rightarrow C_H = 0 \\ \Sigma F_V = 0 \Rightarrow -F + B_V - C_V - q \cdot 4a = 0 \\ \Sigma M_B = 0 \Rightarrow 2aF - 2a \cdot q \cdot 4a - 4a \cdot C_V = 0 \end{cases}$

$\Rightarrow \begin{cases} C_H = 0 \\ C_V = -1\ kN \\ B_V = 13\ kN \end{cases}$

② $\begin{cases} \Sigma F_H = 0 \Rightarrow -C_H + D_H = 0 \\ \Sigma F_V = 0 \Rightarrow C_V + D_V = 0 \\ \Sigma M_D = 0 \Rightarrow M_D - 2a \cdot C_V = 0 \end{cases}$

$\Rightarrow \begin{cases} D_H = 0 \\ D_V = 1\ kN \\ M_D = 2\ kNm \end{cases}$



N [kN]

0

C.

Q [kN]

F ↓

−6 kN

C.

⊖

⊕

7 kN

B V ↑

linear

3.5

0.5

⊖

−1 kN

C.

↑ D V.

Q.

0.25 kNm

M [kNm]

2.

⊕

⊖

0

⊖

L.

↑ M D

−2 kNm